

Engineering Chemistry (CY150); Time: 50 Mins; Minor Exam 1

- 1 Calculate the resonance frequency in units of MHz for an electron and a proton in a magnetic field of 1.2 T. 4 Marks
- 2 Which has the greater energy level separation, a ^{14}N nucleus in an NMR spectrometer operating at 14 T or an electron in an EPR spectrometer operating at 0.30 T? Calculate the energy and prove your answer. 5 Marks
- 3 Calculate the relative population of electron and proton spins in alpha and beta states at the magnetic field of 2.6 T at 25 °C. 5 Marks
- 4 A radical containing three equivalent protons shows a four-line spectrum with an intensity distribution 1:3:3:1. The lines occur at 331.4 mT, 333.6 mT, 335.8 mT, and 338.0 mT. What is the hyperfine coupling constant for each proton in units of Gauss? What is the g-value of the radical in 4 decimal points, given that the spectrometer is operating at 9.332×10^9 Hz? 4 Marks
- 5 The EPR spectrum of a radical with two equivalent nuclei of a particular kind is split into five lines of intensity ratio 1:2:3:2:1. What is the spin of the nuclei? 2 Marks

Physical constant Values:

$$\begin{aligned} Y_e &= 1.761 \times 10^{11} \text{ s}^{-1} \text{ T}^{-1}; \text{ Boltzmann constant} = 1.38 \times 10^{-23} \text{ J/K}; \\ h &= 6.626 \times 10^{-34} \text{ J.s}; \beta_e = 9.274 \times 10^{-24} \text{ J/T}; g_e = 2.0023 \\ Y_{^{14}\text{N}} &= 1.933 \times 10^7 \text{ s}^{-1} \text{ T}^{-1}; Y_p = 2.675 \times 10^8 \text{ s}^{-1} \text{ T}^{-1} \end{aligned}$$



Roll No

25CSE1049

NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

Mid Semester Examinations, October-2025

Course Name: Engineering Chemistry/Chemistry

Date: 11.10.2025

Duration: 1.5 Hours

Course Code: CY 150

Time: 10 PM

Max. Marks: 50

ANSWER ALL QUESTIONS

1. Show the suitable mechanism step-wise for the synthesis of a) teflon, b) poly (vinyl methyl ethers) 6 Marks
2. a) Show the synthesis of two forms of poly isoprene, and b) show the synthesis of styrene butadiene rubber and nitrile rubber. In both the cases (example a and b) compare the Tg values based on the structures of the polymer and comment which has high or low Tg value, and why. 6 Marks
3. Write the synthesis of a) UF resin, b) super glue, and c) conducting polymer 6 Marks
4. Discuss the different classifications based on the arrangement of functional groups on the polymer chain with a proper example. Draw and discuss a method and a mechanism of synthesizing any example of an isotactic polymer 7 Marks
5. $\text{MnSO}_4 + \text{KOH} \longrightarrow \text{A} + \text{B}$ 8 Marks
 $a \text{ A} + \text{O}_2 \longrightarrow 2 \text{ C}$
 $2 \text{ C} + d \text{ H}_2\text{SO}_4 \longrightarrow e \text{ E} + f \text{ F}$
 $e \text{ E} + g \text{ KI} \longrightarrow h \text{ H} + i \text{ I} + j \text{ J}$
Identify the components 'A' to 'J' and the number of moles of the component 'a' to 'j' in the equation above.
6. COD of wastewater shall be calculated by using the formula $\frac{(A-B) \times 8000 \times N}{V}$. Where A and B are the volume of $\text{K}_2\text{Cr}_2\text{O}_7$ consumed in blank and wastewater titration, N is the concentration of $\text{K}_2\text{Cr}_2\text{O}_7$, and V is the volume of the wastewater sample. Write the chemical equations involved in the COD experiment and equate them to arrive at the formula of $\frac{B \times 8000 \times N}{V}$. 10 Marks
7. Calculate the temporary, permanent, and total hardness of a water sample containing $\text{Mg}(\text{HCO}_3)_2 = 7.3 \text{ mg/L}$, $\text{Ca}(\text{HCO}_3)_2 = 16.2 \text{ mg/L}$, $\text{MgCl}_2 = 9.5 \text{ mg/L}$, $\text{CaSO}_4 = 13.6 \text{ mg/L}$, and $\text{CaCl}_2 = 22.2 \text{ mg/L}$ (atomic weight of Mg = 24, Ca = 40, S = 32, Cl = 35.5, respectively) 7 Marks



Roll no 25CSE1049

NATIONAL INSTITUTE OF TECHNOLOGY, GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

End Semester Examinations, December 2025

Course Name: Engineering Chemistry

Date: 11.12.2025

Duration: 3 Hours

Course Code: CY150

Time: 10 AM

Max. Marks: 100

1. Answer all the questions.
2. Answers for sub-questions have to be labelled properly
3. All calculations/workouts must be shown in the answer script properly

- 1 ✓ Classify the polymers based on the intermolecular forces. Explain the classification and type of forces intermolecular forces and their influence on the properties of the polymer. Give two examples of synthesis of polymer for each classification 12 Marks
- 2 ✓ Write the mechanism of living polymerization with a proper example of a polymer that can be synthesized by this method 6 Marks
- 3 ✓ Explain and discuss the influence of intermolecular forces on the thermal properties of any two polymers with proper glass transition values. 4 Marks
- 4 ✓ Show the synthesis of two geometrical isomers of a polymer. Explain the effect of these isomeric structures on the physical properties of polymers. 3 Marks
- 5 ✓ The following is the data obtained by the titration of the mixture of A (oil) and B (water) with C (surfactant). Calculate the % A, B, and C, tabulate them, and mark these points on the phase diagram. Draw a binodal curve and mark the plait point on the phase diagram. 12 Marks

Point	A (mL) (oil)	B (mL) (water)	C(mL) (surfactant)
i	6	2	2
j	8	1	1
k	3	5	2
l	2	4	4
m	1	6	3
n	4	3	3
o	3	3	4
p	5	4	1
q	7	1	2

- 6 ✓ Draw, calculate degrees of freedom, and explain the phase diagram of a system involving more than one triple point. Compare this phase diagram with any system involving only one triple point and explain the degrees of freedom for the same. Explain the reason for seeing more triple points in any system. 8 Marks

- 7 Draw the phase diagrams involving a peritectic point. Calculate all the degrees of freedom. 5 Marks
- 8 Zinc(II) and the ligand L form a 1:1 complex that absorbs strongly at 600 nm. As long as the molar concentration of L exceeds that of zinc(II) by a factor of 5, the absorbance depends only on the cation concentration. Neither zinc(II) nor L absorbs at 600 nm. A solution that is 1.59×10^{-4} M in zinc(II) and 1.00×10^{-3} M in L has an absorbance of 0.352 in a 1.00 cm cell at 600 nm. Calculate 6 Marks
- (a) The percent transmittance of this solution.
 (b) The percent transmittance of this solution in a 2.50 cm cell,
 (c) the molar absorptivity of the complex.
- 9 When measured with a 1.00 cm cell, a 7.50×10^{-5} M solution of species A exhibited absorbances of 0.155 and 0.755 at 475 and 700 nm, respectively. A 4.25×10^{-5} M solution of species B gave absorbances of 0.702 and 0.091 under the same circumstances. Calculate the concentrations of A and B in the solutions that yielded the following absorbance data in a 2.50 cm cell: (a) 0.439 at 475 nm and 1.025 at 700 nm 8 Marks
- 10 Gaseous HCl exhibits an IR absorption at 2890 cm^{-1} due to the hydrogen-chlorine stretching vibration. 8 Marks
- (a) Calculate the force constant for the bond.
 (b) Calculate the wavenumber of the absorption band for DCl, assuming the force constant is the same as that calculated in part (a).
 Note that the MW of Cl is 35.5 g/mol
- 11 How many vibrational modes will the SO_2 molecule have? How many IR absorption bands would sulfur dioxide be expected to have? Identify the vibrational modes responsible for IR bands. 7 Marks
- 12 A nucleus has a spin quantum number of $5/2$. How many magnetic energy states does this nucleus have? What is the magnetic quantum number of each? Calculate the length of the angular momentum and the magnetic moment vector. 4 Marks
- 13 In the room-temperature ^1H NMR spectrum of methanol, no spin-spin coupling is observed. But when a methanol sample is cooled to -41°C , the exchange rate of the hydroxyl proton slows sufficiently so that splitting is observed. Sketch spectra for methanol at the two temperatures. 4 Marks
- 14 Describe the cold and hot soda lime process of water treatment with appropriate chemical reactions. List out at least 4 advantages of the hot soda lime process over the cold soda lime process. 7 Marks
- 15 Explain corrosion due to oxygen and how it can be overcome with an appropriate chemical reaction. 4 Marks
- 16 Explain the carbonate conditioning and phosphate conditioning with appropriate chemical reactions. 2 Marks

****ALL THE BEST****



NATIONAL INSTITUTE OF TECHNOLOGY GOA
Cuncolim, Salcete, South Goa-403703
Programme Name: B.Tech.
End Semester Examinations, April 2026

Course Name: Engineering Chemistry
Date: 29.04.2026
Duration: 3 Hours

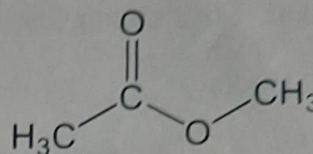
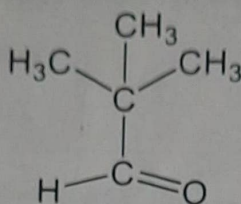
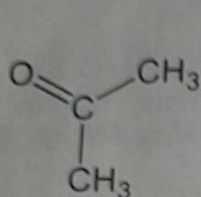
Course Code: CY150
Time: 10 AM
Max. Marks: 100

1. Answer all the questions.
2. Answers for sub-questions have to be labeled properly
3. All calculations/workouts must be shown in the answer script properly

- 1 Write the synthesis of a conducting polymer and a fiber polymer. 6 Marks
- 2 Write a suitable mechanism of synthesis of polyacrylonitrile and poly methyl methacrylate. 6 Marks
- 3 How bulk polymerization is different from solution polymerization. Tabulate and compare with examples of polymers synthesized. 5 Marks
- 4 Discuss the tacticity in a polymer with examples and comment on the stereoregularity. 4 Marks
- 5 Compare and state the expected influence of structural differences in a cis and trans polymer, and show the synthetic step and structure of these polymers. 4 Marks
- 6 Compare the sublimation behaviour in two different one-component systems. Explain what you observe for each system during sublimation and comment on each. Draw the phase diagram and derive the degrees of freedom of all the points, curves, and areas of both. 6 Marks
- 7 Derive the Gibbs phase rule stepwise. Explain a situation where degrees of freedom become negative, and comment on whether such F values exist 6 Marks
- 8 Draw and explain the phase diagram of the system involving a peritectic point during the phase transition 6 Marks
- 9 Draw a phase diagram of a ternary system consisting of your choice of components. Tabulate all the values neatly, and show stepwise calculation of each point, labeling them as A, B, C, D, and E on the table and phase diagram. Draw the binodal curve, tie line, and identify the plait point. State the application of this phase diagram. 7 Marks
- 10 Calculate the population ratio of spins in their spin states when protons are placed in a homogenous magnetic field of (a) 0.30 T, and (b) 1.5 T at 25°C. 4 Marks
- 11 Calculate the resonance frequency of a proton in the units of MHz when the spectrometer was operated at the magnetic fields of 2.5 T and 12.5 T. 4 Marks

12 Predict the low and high-resolution NMR spectra of the following molecules

9 Marks



- 13 A proton resonates at 7.0×10^{-6} T downfield from TMS in an NMR spectrometer operating at 2.5 T. Calculate the chemical shift of the proton in ppm. 2 Marks
- 14 How many vibrational modes will SO_2 and HCN molecules have? Identify the modes. How many IR absorption bands would this molecule be expected to have, and justify? 12 Marks
- 15 A 2.5 ml solution containing 7.9 ppm of Fe(III) is treated with an appropriate excess of KSCN to form the Fe(SCN)^{2+} complex. The Fe(SCN)^{2+} complex has an absorption maximum at 580 nm and a molar absorptivity of $7.0 \times 10^3 \text{ L cm}^{-1} \text{M}^{-1}$. This solution is diluted to 50 ml. What is the absorbance of the resulting solution at 580 nm in a 2.5 cm cell? Calculate the transmittance percentage. 6 Marks
- 16 The organic compound, $\text{CH}_3\text{CH}=\text{CHCHO}$ has a strong absorption in the ultraviolet at 213 nm and a weak absorption at 333 nm. Justify these features and assign the ultraviolet absorption transitions. 4 Marks
- 17 Describe the cold soda lime process with appropriate chemical equations and the diagram. 4 Marks
- 18 A one-liter water sample was tested for minerals present, and observed to have 18 mg of $\text{Ca(HCO}_3)_2$, 17 mg of CaSO_4 , 12 mg of CaCl_2 and 14 mg of MgCl_2 . Calculate the temporary and permanent hardness of the water sample. 5 Marks

Physical Constants

1. Plank constant = $6.626 \times 10^{-34} \text{ J.s}$
2. Atomic mass of Fe: 55.9 amu
3. MW (g/mole): $\text{Ca(HCO}_3)_2 = 162$, $\text{CaSO}_4 = 136$, $\text{CaCl}_2 = 111$, $\text{MgCl}_2 = 95$
4. Gyromagnetic ratio of proton: $2.675 \times 10^8 \text{ s}^{-1} \cdot \text{T}^{-1}$
5. Boltzmann constant: $1.381 \times 10^{-23} \text{ J.K}^{-1}$

****ALL THE BEST****